PRD: SMB Risk Verification (Quick check)

Problem

Small-business insurance applications are self-declared, and misclassification is common (Verisk: about 52% of small commercial records carry the wrong industry code). The evidence to check them is scattered across public registers, city records and the business's own website.

Goal and users

Show that we can verify an underwriter's application answers by joining messy public sources, reading unstructured text with an LLM, and showing the evidence behind every answer. Users: underwriters (check answers) and carrier or MGA demo audiences (see accuracy on real data). **Not in scope:** loss prediction, automated decisions, storing Google reviews, lines other than restaurants. **Scope:** Manhattan food businesses listed in NYC health inspections; configurable to all five boroughs.

Source map: what each source actually reports

"States" means the source records the fact. "Inferred" means we derive it, and the evidence shown says how. Every dataset link below was checked and resolves to the named dataset.

Source	Link	Fields we use	What we derive	How	Level
NYC health inspections	43nn-pn8j	name, address, cuisine, inspection date, action, violation, critical flag, grade, BIN, BBL	Business identity; first inspection; last grade; closures and critical violations in 3 yrs	States	Business
NYS liquor licences	9s3h-dpkz	licence type, legal name, trade name, address, location, first issue, expiry	**Serves alcohol = yes**; licence first issued; legal name	States (licence held). No licence found = weak "no"	Business
NYC Dining Out licences	fpeh-f7ci	legal name, trade name, sidewalk/roadway type, status, issue date	**Outdoor seating = yes**	States (licensed sidewalk or roadway dining)	Business
NYS company register	n9v6-gdp6	entity name, formation date, entity type	Legal entity formed	States	Legal entity
FDNY summonses (OATH)	jz4z-kudi	respondent name, block/lot, charge, hearing result	Number of fire summonses and main charges	States; attributed only when respondent name matches the business	Business
PLUTO	64uk-42ks	year built, floors, building class	Building age, height, class	States	Tax lot
DOB violations	3h2n-5cm9	violation status, description	Open DOB violations	States	**Whole building**, not the tenant
DOB ECB violations	6bgk-3dad	violation status, description	Active ECB violations	States	**Whole building**, not the tenant
311 complaints	erm2-nwe9	complaint type, venue type, time	**Open after midnight = yes** if 2+ bar/restaurant noise complaints between 00:00 and 05:00	Inferred (complaint time is not opening hours); used only where one food business is on the lot	Tax lot
OpenStreetMap	licence · Overpass API	venue category, opening hours, outdoor_seating, delivery, website	**Serves alcohol = yes** only if mapped as bar, pub, nightclub or beer garden; open after midnight from posted hours; outdoor seating and delivery tags; website URL	Category is inferred (OSM has no alcohol field); hours and tags are crowd-entered	Venue

Source	Link	Fields we use	What we derive	How	Level
Business website	No dataset: each business's own site (URL from its OpenStreetMap entry), shown on its profile	home page and one menu page	Serves alcohol (drinks or wine list); **fried items on menu**; delivery (own or app); posted hours past midnight; outdoor seating mention	Inferred from what the business publishes. Read by an LLM (Gemini 2.5 Flash), which must copy the supporting sentence word for word from the page. Deep frying comes from menu items (fries, tempura, wings), not a statement	Business
Google Maps reviews (live)	No dataset: a Google Maps place link is returned with each answer · method	Maps place data and reviews, summarised live by Gemini	Nothing stored or derived; shown to the underwriter only	n/a	Venue

Requirements

1 Download each source as one snapshot; fail if the row count differs from the server.
2 Match businesses across sources; keep the method and score of every match.
3 For each website answer, Gemini must copy the sentence from the page that supports it. We then search the page for that exact sentence; if it is not there, the answer is discarded. This stops the LLM from inventing evidence.
4 Resolve facts by trust (official > own website > missing licence > crowd/complaints) and flag conflicts.
5 App: search, profile with evidence links, application check (consistent / contradicted / no evidence), live Maps insight.

Success metrics

Metric	How it is measured
Accuracy of website answers	Website "serves alcohol" answers compared with the liquor licence records; disagreements hand-reviewed
Evidence integrity	Share of website answers whose supporting sentence is found word for word in the page
Match accuracy	Hand-check of a random sample of matches for each matching method
Coverage	Share of businesses with a value for each fact, and linked to each source
Conflicts surfaced	Number of businesses where sources disagree on an application fact

Constraints, risks, next steps

- Google Maps terms forbid saving reviews or deriving content from them; NYC Open Data needs legal review for commercial use.
- Building-level records cannot be pinned to one tenant where several businesses share the lot.
- Public data shows facts, not losses. Next: read websites for every business, extend to five boroughs, and test against a design partner's real applications and claims.